

THESE P AND I DIAGRAM IS DIAGRAMATIC ONLY. DESIGN OF PIPE LINES MUST BE INVESTIGATED FOR VENTING OF GAS AND VAPOR POCKETS IN PIPING AND EQUIPMENT. LOW POINTS IN PIPING, PUMPS AND EQUIPMENT FOR FREEZING AND DRAINING, AND ACCESSIBILITY OF ALL VALVES, FLANGES, INSTRUMENTS INCLUDING THERMOCOUPLES, ETC.

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14780-8120-25-23-0001F

B

C

D

E

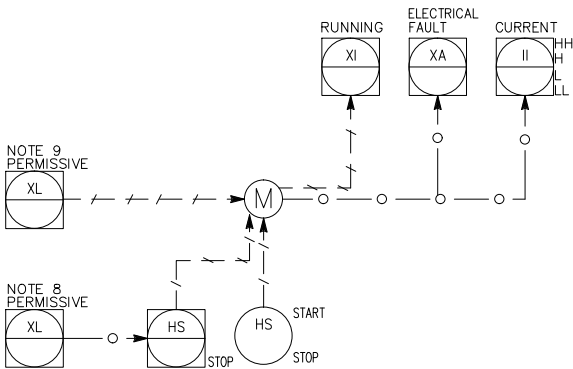
F

G

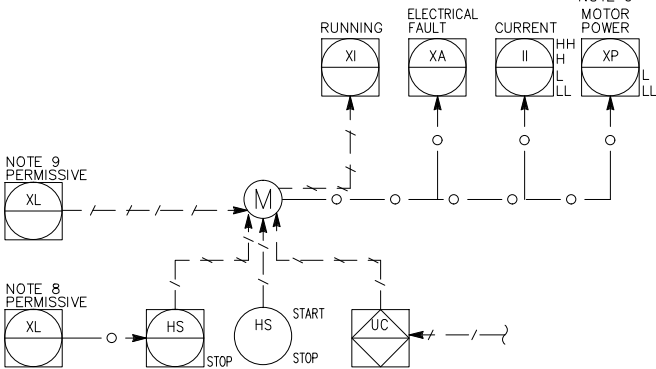
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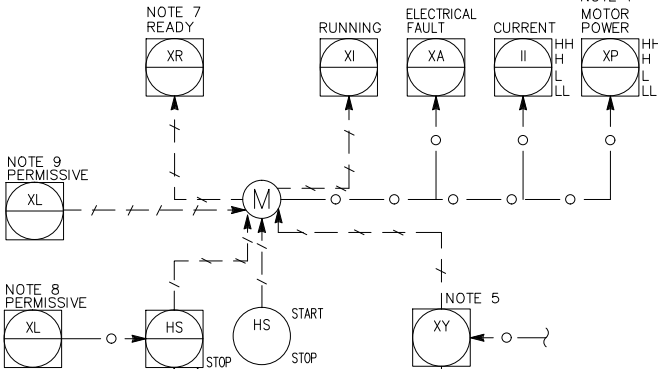
TYPICAL PUMP MOTOR CONTROL SCHEMATIC



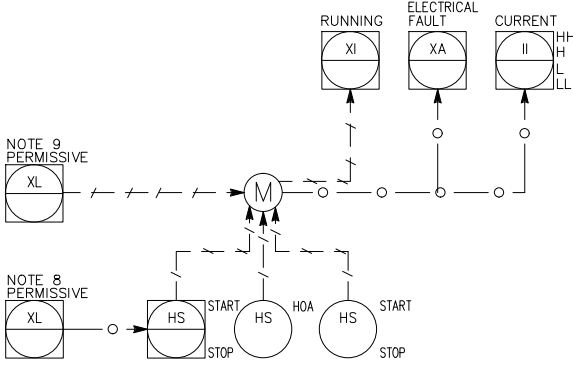
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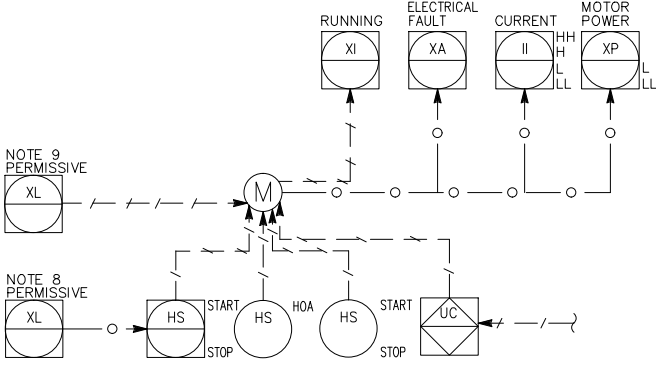
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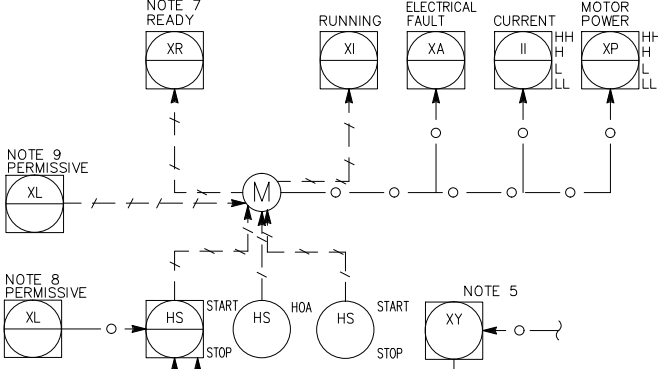
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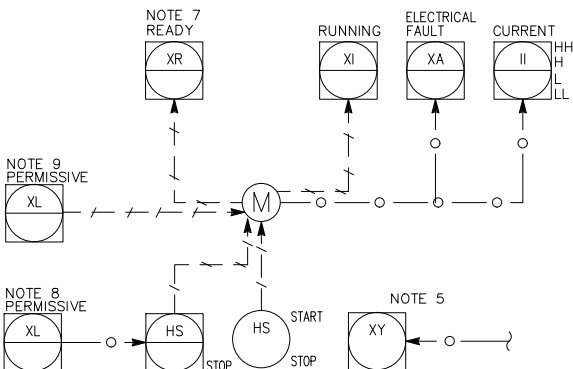
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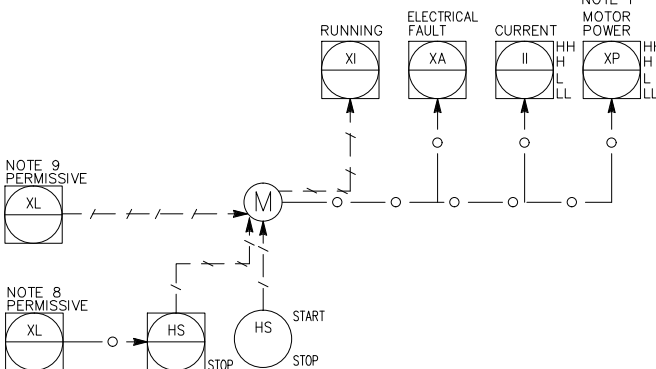
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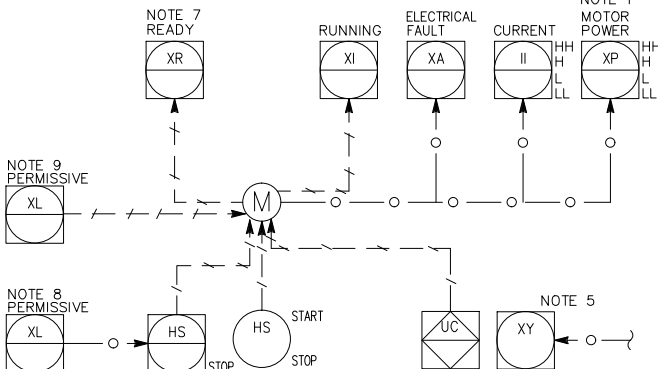
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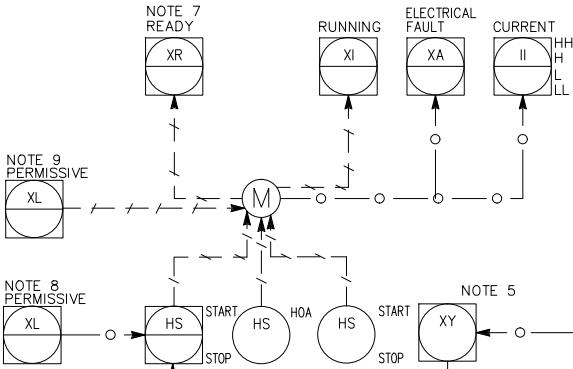
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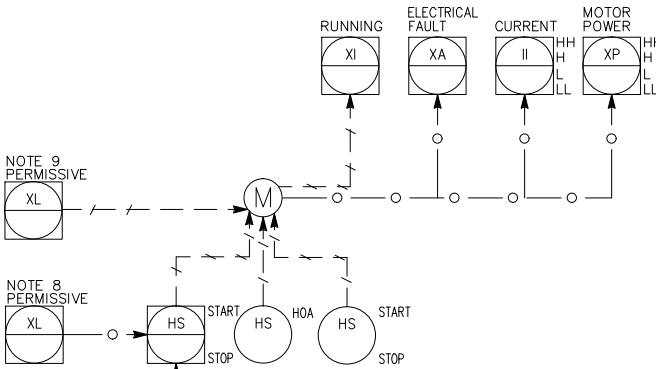
TYPE G



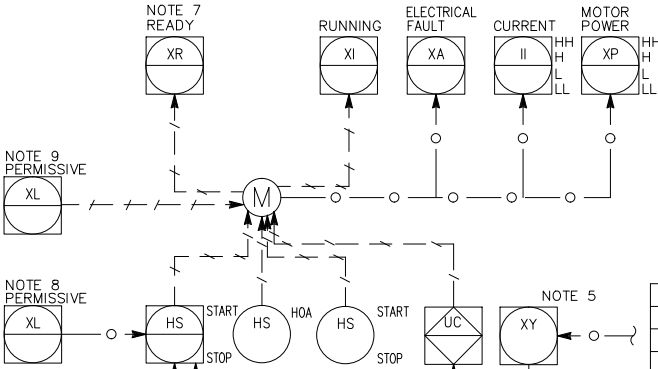
TYPE K



TYPE D



TYPE H



TYPE L

|      |                                    |     |      |       |      |          |
|------|------------------------------------|-----|------|-------|------|----------|
| Z1   | AS BUILT                           | HSO | SWP  | JSC   | SWL  | 4/25/16  |
| S4   | ISSUED FOR CONSTRUCTION NO HOLDS   | HSO | SWP  | JSC   | SWL  | 10/15/14 |
| S3   | ISSUED FOR CONSTRUCTION NO HOLDS   | HSO | SWP  | JSC   | SWL  | 10/8/14  |
| S2   | ISSUED FOR CONSTRUCTION NO HOLDS   | HSO | SWP  | JSC   | SWL  | 9/16/14  |
| S1   | ISSUED FOR CONSTRUCTION NO HOLDS   | HSO | SWP  | JSC   | SWL  | 8/20/14  |
| R2   | ISSUED FOR CONSTRUCTION WITH HOLDS | HSO | SWP  | JSC   | SWL  | 4/4/14   |
| R1   | ISSUED FOR CONSTRUCTION WITH HOLDS | HSO | SWP  | JSC   | SWL  | 2/28/14  |
| A2   | APPROVED FOR DESIGN                | HSO | SWP  | JSC   | SWL  | 2/7/14   |
| REV. | DESCRIPTION                        | BY  | CHK. | APPR. | DATE |          |

GENERAL NOTES

NOTES

1. HIGH AND LOW MOTOR POWER ALARMS AND SHUTDOWNS ARE PROVIDED FOR MAGNETIC SEALESS PUMPS.
2. SHUTDOWN CIRCUITRY SHALL BE DESIGNED TO STOP THE OPERATING PUMP FOR AN ABNORMAL CONDITION BY ANY OF THE LISTED SHUTDOWN VARIABLES.
3. ALARM CIRCUITRY SHALL BE DESIGNED TO SERIES CONNECT ALL LISTED ALARM CONTACTS, FOR EACH PUMP SERVICE, TO ACTIVATE THE RELATED ALARM IN THE CONTROL CENTER.
4. DELETED.
5. XY CAN BE FY, LY, PY AND UY.
6. LOW LOW MOTOR POWER ALARM AND SHUTDOWN ARE PROVIDED FOR P-2102A/B AND P-2203A/B
7. READY SIGNAL IS APPLIED ONLY FOR AUTO-START PUMPS.
8. PERMISSIVE SIGNAL FOR LV MOTOR IS COMBINED TO DCS STOP.  
IN ADDITION, PERMISSIVE SIGNAL FROM PROCESS CONTROL LOGIC IS CONNETED TO THIS PERMISSIVE SIGNAL IF REQUIRED.
9. PERMISSIVE SIGNAL FOR HV MOTOR IS CONNECTED TO MCC DIRECTLY VIA HARD WIRE SIGNAL.  
IN ADDITION, PERMISSIVE SIGNAL FROM PROCESS CONTROL LOGIC IS CONNETED TO THIS PERMISSIVE SIGNAL IF REQUIRED.
10. TEMPERATURE SWITCH IS PROVIDED BY MAGNETIC PUMP VENDOR AND TEMP. HIGH SIGNAL WILL TRIP MAGNETIC PUMP.



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BASED ON

PIPING AND INSTRUMENT DIAGRAM  
TYPICAL PUMP MOTOR CONTROL SCHEMATIC (1/2)  
CONCENTRATION, DECOMPOSITION, AND  
NEUTRALIZATION SECTION  
UOP PHENOL PROCESS UNIT

|           |                        |      |       |      |       |
|-----------|------------------------|------|-------|------|-------|
| BY        | DATE                   | DWG. | DSGN. | CHK. | APPR. |
|           |                        |      |       |      |       |
| PROJ. NO. | DWG. NO.               | REV. |       |      |       |
| PEN       | 120117                 |      |       |      | Z1    |
| PPCL      | 14780-8120-25-23-0001F |      |       |      |       |

SCALE